

# Lecture 10: Distributions

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## 1 Introduction

How far can we push weak solutions? We have defined a weak derivative, which may go well beyond just Sobolev spaces, and we have also observed that there are some interesting objects that behave like measures. Our case in point is the heat kernel  $H_t(x)$ , which converged pointwise as  $t \rightarrow 0$  to

$$\begin{cases} \infty & x = 0 \\ 0 & \text{else} \end{cases}$$

but when we measured it via integration,  $\lim_{t \rightarrow 0} \int H_t(y) f(x - y) dy = f(x)$ , so this limit acts like the probability distribution assigning probability 1 to 0 and probability 0 to everything else. We noticed before that  $H_t$  looked like a derivative of a smoothed-out step function, so we are saying that this “point density” looks like the derivative of the step function (which is not differentiable, so we may try using weak definitions).

Surely, the weird pointwise behavior isn’t enough to describe this, because we can generate other functions with the same pointwise behavior where this fails. We can see this behavior arise in another way physically, in electrostatics, through Coulomb’s law.

### 1.1 Model: Coulomb’s Law

Charles-Augustin de Coulomb published an empirical observation of electrostatics in the 18th century, describing that a particle with electric charges  $q_0$  located at the origin generates an electric field given by

$$E(x) = \frac{kq_0 x}{|x|^3}$$

for a constant  $k$ . Recall that Gauss’ law also told us  $\nabla \cdot E = 4\pi k\rho$ . The weird part about this condition is that, except at  $x = 0$ ,

$$\nabla \cdot \frac{x}{|x|^3} = \frac{\nabla \cdot x}{|x|^3} - \frac{3x}{|x|^4} \nabla |x| = 0$$

so the behavior on a null set must somehow cause Gauss’ law. We rectify this by using the weak form of Gauss’ law,

$$\int_{\mathbb{R}^3} E \cdot \nabla \psi dx = -4\pi k \int_{\mathbb{R}^3} \rho \psi dx$$

for all test functions  $\psi$ . As long as we exclude the origin, we can integrate by parts, so we write

$$\begin{aligned} \int_{\mathbb{R}^3} E \cdot \nabla \psi dx &= \lim_{\epsilon \rightarrow 0} \int_{|x| \geq \epsilon} E \cdot \nabla \psi dx \\ &= \lim_{\epsilon \rightarrow 0} \int_{|x|=\epsilon} \nu \cdot E \psi dS = - \int_{|x|=\epsilon} \frac{kq_0}{\epsilon^2} \psi dS \\ &= -4\pi kq_0 \psi(0) \end{aligned}$$

so this requires

$$\int_{\mathbb{R}^3} \rho \psi dx = q_0 \psi(0)$$

another instance of a point density.

## 2 Definitions, Convergence, Derivatives

The theory of distributions was developed independently in the mid-20th century by Sergei Sobolev and Laurent Schwartz, as the concept of the dual to the test functions.

A *distribution* on a domain  $\Omega \subset \mathbb{R}^n$  is a linear function  $\mathbb{C}_c^\infty(\Omega) \rightarrow \mathbb{C}$  that is continuous. We write the function as  $u$ , and write evaluations as pairings  $(u, \psi)$ .

By continuity here, we mean that if  $\{\psi_n\} \rightarrow \psi$  in  $C_c^\infty$ , then  $(u, \psi_n) \rightarrow (u, \psi)$ , where converging in  $C_c^\infty(\Omega)$  means that there exists some compact  $K$  so all the  $\psi_n$  are supported in  $K$  and all of their partial derivatives converge uniformly to those of  $\psi$  on  $K$ .

We denote the space of distributions by  $D'(\Omega)$  (the prime signifying dual). Notice that  $L_{loc}^1(\Omega) \subset D'(\Omega)$  via

$$(f, \psi) = \int_{\Omega} f \psi dx$$

and we have also seen the dirac delta  $(\delta_x, \psi) = \psi(x)$ .

We may place the *weak topology* on  $D'(\Omega)$ , by saying  $u_k \rightarrow u$  in  $D'(\Omega)$  if

$$\lim_{k \rightarrow \infty} (u_k, \psi) = (u, \psi)$$

for all  $\psi \in C_c^\infty$ . We may finally formalize the idea behind the heat kernel:

**[2.1]** Given  $f \in L^1(\mathbb{R}^n)$  satisfying  $\int_{\mathbb{R}^n} f dx = 1$ , define  $f_a(x) = a^n f(ax)$ . Then,

$$\lim_{a \rightarrow \infty} f_a = \delta_0$$

as a distributional limit.

*Proof.* For  $\psi \in C_c^\infty(\mathbb{R}^n)$ ,

$$(f_a, \psi) = \int_{\mathbb{R}^n} a^n f(ax) \psi dx = \int_{\mathbb{R}^n} f(x) \psi(x/a) dx$$

so

$$|(f_a, \psi) - \psi(0)| \leq \int_{\mathbb{R}^n} |f(x)(\psi(x/a) - \psi(0))| dx$$

We now use the decay of  $f$  and the continuity of  $\psi$ . Pick  $\epsilon > 0$ . Assume  $|\psi| \leq M$ . Then, we may find  $R$  so

$$\int_{|x| > R} |f| dx \leq \frac{\epsilon}{2M}$$

and  $\delta > 0$  so  $|\psi(y) - \psi(0)| < \epsilon/2$  for  $|y| < \delta$ . Setting  $a \geq R/\delta$ ,

$$\int_{\mathbb{R}^n} |f(x)(\psi(x/a) - \psi(0))| dx \leq \int_{|x| > R} |f|(2M) dx + \int_{|x| \leq R} |f| \epsilon/2 dx < \epsilon$$

so  $(f_a, \psi) \rightarrow \psi(0)$  as desired. □

We may also extend many operations to  $D'$ . Because the idea of the integral pairing is so central to the construction of the distributions, we define many operations by mimicking that case. For example, when  $f \in C^\infty(\Omega)$ , we define  $uf$  to be the distribution so

$$(uf, \psi) = (u, f\psi)$$

for all  $\psi$ . Similarly, we define the *distributional derivative*  $\partial^\alpha u$  to be the distribution

$$(\partial^\alpha u, \psi) = (u, (-1)^{|\alpha|} \partial^\alpha \psi)$$

**Example:** Notice that

$$\int_{\mathbb{R}} -1_{[0, \infty)} \psi' dx = \int_0^\infty -\psi'(x) dx = \psi(0)$$

so that

$$\frac{d}{dx}(-1_{[0, \infty)}) = \delta_0$$

as distributions. We may compute further derivatives by noticing

$$(\partial^\alpha \delta_0, \psi) = (-1)^{|\alpha|} \partial^\alpha \psi(0)$$

**Example:**

The function  $\ln|x|$  is locally integrable and so it defines a distribution on  $\mathbb{R}$ . We may expect the distributional derivative to be  $\frac{1}{x}$  for  $x \neq 0$ , but this is not locally integrable. To understand this, we go back to the integral pairing:

$$\begin{aligned} \int_{\mathbb{R}} -\psi' \ln|x| dx &:= \lim_{\epsilon \rightarrow 0} - \int_{|x| \geq \epsilon} \psi' \ln|x| dx \\ &= \lim_{\epsilon \rightarrow 0} \int_{|x| \geq \epsilon} \frac{\psi(x)}{x} dx + [\psi(\epsilon) - \psi(-\epsilon)] \ln(\epsilon) \\ &= \lim_{\epsilon \rightarrow 0} \int_{|x| \geq \epsilon} \frac{\psi(x)}{x} dx \end{aligned}$$

since

$$\lim_{\epsilon \rightarrow 0} [\psi(\epsilon) - \psi(-\epsilon)] \ln(\epsilon) = \psi'(0) \lim_{\epsilon \rightarrow 0} 2\epsilon \ln|\epsilon| = 0$$

The limit definition of the integral is given a special name, the principal value distribution  $PV[x^{-1}]$ , and only converges because it is taken symmetrically.

**Example:**

Gauss' law from the introduction, combined with Coulomb's law, may be rewritten as

$$\nabla \cdot \frac{x}{|x|^3} = 4\pi\delta$$

### 3 Fundamental Solutions

Imagine we have a matrix  $L$ , and vectors  $f_i$  so  $Lf_i = e_i$ . How do we solve  $Lf = v$ ? We superimpose solutions  $f = \sum v_i f_i$  so  $Lf = \sum v_i e_i = v$ .

Now, if instead we have a differential operator  $L$  and solutions  $\Phi_x$  so  $L\Phi = \delta_0$ , we can solve  $Lu = f$  by superimposing solutions

$$u = \int \Phi(x - y)f(y)dy$$

Then, as long as we are justified in swapping the integral and derivatives,

$$Lu = \int L\Phi(x - y)f(y)dy = f(x)$$

as desired. We therefore call the solution  $\Phi_x$  the *fundamental solution* at  $x$ .

We're treating the fundamental solution like a function here, but that may not always be so. To justify these actions, we need to consider convolutions with distributions and how distributional derivatives work with convolutions.

In the first case, when  $f, g \in L^1_{loc}(\mathbb{R}^n)$  and  $\psi \in C_c^\infty(\Omega)$ ,

$$(f * g, \psi) = \int \int f(y)g(x - y)\psi(x)dydx = \int f(y) \int g(y - x)\psi(x)dx dy$$

so that we may move the convolution off  $f$  and onto  $\psi$  by taking  $g^-(x) = g(-x)$ . Then,

$$(f * g, \psi) = (f, g^- * \psi)$$

Using the right hand side, we may define the convolution of  $u \in D'(\mathbb{R}^n)$  and a test function  $\phi \in C_c^\infty(\mathbb{R}^n)$  by

$$(u * \phi, \psi) = (f, \phi^- * \psi)$$

for all  $\psi \in C_c^\infty(\mathbb{R}^n)$ . In particular, the  $\delta_0$  distribution gives

$$(\delta_0 * \phi, \psi) = \phi^- * \psi(0) = (\phi, \psi)$$

so  $\delta_0 * \phi = \phi$ .

For the second complication, we have the following

**[3.1]** For  $w \in D'(\mathbb{R}^n)$  and  $\psi \in C_c^\infty(\mathbb{R}^n)$ ,

$$D^\alpha(w * \psi) = (D^\alpha w) * \psi = w * (D^\alpha \psi)$$

*Proof:*

For  $\phi, \psi$  both test functions,

$$\begin{aligned} D^\alpha(\phi * \psi) &= D^\alpha \int_{\mathbb{R}^n} \phi(x - y)\psi(y)dy \\ &= \int_{\mathbb{R}^n} D^\alpha \phi(x - y)\psi(y)dy = (D^\alpha \phi) * \psi \end{aligned}$$

as desired. Next, for  $w \in D'(\mathbb{R}^n)$  and  $\phi, \psi \in C_c^\infty(\mathbb{R}^n)$ ,

$$\begin{aligned} (D^\alpha(w * \phi), \psi) &= (-1)^{|\alpha|}(w * \phi, D^\alpha \psi) = (-1)^\alpha(u, \phi^- * D^\alpha \psi) = (-1)^\alpha(u, D^\alpha(\phi^- * \psi)) \\ &= ((D^\alpha u), \phi^- * \psi) = ((D^\alpha u) * \phi, \psi) \end{aligned}$$

or

$$(-1)^\alpha(u, D^\alpha(\phi^- * \psi)) = (u, D^\alpha \phi^- * \psi) = (u * D^\alpha \phi, \psi)$$

as desired.

*Q.E.D.*

We now formally state our heuristic

**[3.2]** If  $L$  is a constant coefficient operator on  $\mathbb{R}^n$  with fundamental solution  $\Phi \in D'(\mathbb{R}^n)$  satisfying  $L\Phi = \delta_0$ , then for  $f \in C_c^\infty(\mathbb{R}^n)$ ,

$$Lu = f$$

is solved by

$$u = \Phi * f$$

*Proof:*

Let

$$Lf = \sum_{|\alpha| \leq m} a_\alpha D^\alpha f$$

Then,

$$L(\Phi * f) = \sum_{|\alpha| \leq m} a_\alpha D^\alpha (\Phi * f) = (L\Phi) * f = \delta_0 * f = f$$

*Q.E.D.*

### 3.1 Laplace Solution

To motivate our fundamental solution, we recall that the Laplacian is rotationally invariant. It may then make sense to try to reduce to an ODE in  $r$ , in which case we have

$$r^{1-n} \partial_r (r^{n-1} \partial_r \rho) = \partial_r^2 \rho + \frac{n-1}{r} \partial_r \rho = \delta_0$$

From Coulomb's law in the introduction, we saw that  $\frac{1}{C r^2}$  gives a solution in 3D for the right constant. We then may try a general monomial  $r^{-l}$ , in which case

$$(-l)(-l-1)r^{-l-2} + (n-1)(-l)r^{-l-2}$$

is only zero for  $r \neq 0$  if  $l = n-1$ . We then must check at  $r = 0$ :

**[3.3]** On  $\mathbb{R}^n$ , the operator  $-\Delta$  has the fundamental solution

$$\Phi(x) = \begin{cases} -\frac{1}{2\pi} \ln(r) & n = 2 \\ \frac{1}{(n-2)A_n r^{n-2}} & n \geq 3 \end{cases}$$

for  $A_n = \text{vol}(B(0,1))$  in  $\mathbb{R}^n$ .

*Proof:*

We wish to show

$$(-\Delta \Phi, \psi) = \psi(0)$$

First, we notice that

$$\nabla \Phi(x) = -\frac{x}{A_n r^n}$$

so

$$\begin{aligned}
 \int_{\mathbb{R}^n} \Delta \Phi \psi dx &= - \int_{\mathbb{R}^n} \nabla \Phi \cdot \nabla \psi dx \\
 &= \frac{1}{A_n} \int_{\mathbb{R}^n} \frac{x}{r^n} \cdot \nabla \psi dx \\
 &= \frac{1}{A_n} \int_{\mathbb{R}^n} \frac{1}{r^{n-1}} \partial_r \psi dx \\
 &= \frac{1}{A_n} \int_{S^{n-1}} \int_0^\infty \partial_r \psi dr dS = -\psi(0)
 \end{aligned}$$

as desired.

*Q.E.D.*

When the domain is not  $\mathbb{R}^n$ , it becomes a bit harder to use this symmetry to our advantage. George Green developed a method for doing so in 1828, by connecting fundamental solutions to boundary value problems through an integral formula.

Let  $\Phi$  be the fundamental solution of the Laplacian on  $\mathbb{R}^n$  given above. Let  $\Phi_y(x) = \Phi(x - y)$ .

**[3.4] Green's Representation Formula** Suppose that  $\Omega \subset \mathbb{R}^n$  is a bounded domain with piecewise  $C^1$  boundary. For  $u \in C^2(\bar{\Omega})$ ,

$$u(y) = - \int_{\Omega} \Phi_y \Delta u dx + \int_{\partial\Omega} [\Phi_y \partial_\eta u - u \partial_\eta \Phi_y] dS$$

for  $y \in \Omega$ .

*Proof:*

Because the point  $y$  is fixed, we assume  $y = 0$ . Pick some  $\epsilon > 0$  small enough so  $\overline{B(0, \epsilon)} \subset \Omega$ . Then, on  $\bar{\Omega} - B(0, \epsilon)$ ,  $\Phi$  is smooth and has  $\Delta \Phi = 0$ . We then have

$$\begin{aligned}
 \int_{\Omega - \overline{B(0, \epsilon)}} \Phi \Delta u dx &= \int_{\partial\Omega} (\Phi \partial_\eta u - u \partial_\eta \Phi) dS + \\
 &\quad \int_{\partial B(0, \epsilon)} (\Phi \partial_\eta u - u \partial_\eta \Phi) dS
 \end{aligned}$$

Because  $\Delta u$  is continuous and  $\Phi$  is locally integrable,

$$\lim_{\epsilon \rightarrow 0} \int_{\Omega - \overline{B(0, \epsilon)}} \Phi \Delta u dx = \int_{\Omega} \Phi \Delta u dx$$

so to prove the representation formula, we need to show

$$\lim_{\epsilon \rightarrow 0} \int_{\partial B(0, \epsilon)} (\Phi \partial_\eta u - u \partial_\eta \Phi) dS = u(0)$$

We treat the two terms separately. First,  $\nabla u$  is bounded on  $\bar{\Omega}$ , so we take

$$|\partial_\eta u| \leq |\nabla u| \leq M$$

so

$$\left| \int_{\partial B(0, \epsilon)} \Phi \partial_\eta u dS \right| \leq \frac{M}{n-1} \epsilon$$

(or  $M\epsilon \ln(\epsilon)$  if  $n = 1$ ), which disappears as  $\epsilon \rightarrow 0$ .

However,  $\partial_\eta \Phi = \partial_r \Phi = -\frac{1}{A_n r^{n-1}}$  implies

$$-\int_{\partial B(0,\epsilon)} u \partial_r \Phi dS = \frac{1}{|\partial B(0,\epsilon)|} \int_{\partial B(0,\epsilon)} u dS$$

which converges to  $u(0)$  as  $\epsilon \rightarrow 0$  by continuity.

*Q.E.D.*

The original goal of this representation formula was to solve the Poisson problem with inhomogeneous Dirichlet boundary condition. Suppose there exists a family of functions  $H_y \in C^2(\overline{\Omega})$  for  $y \in \Omega$  such that

$$\begin{aligned} \Delta H_y &= 0 \\ H_y|_{\partial\Omega} &= \Phi_y|_{\partial\Omega} \end{aligned}$$

Then,  $G_y = \Phi_y - H_y$  has homogeneous boundary conditions and has  $\Delta G_y = \delta_y$  in  $\Omega$ . This is called the Green's function for  $\Omega$ .

**[3.5]** Suppose  $\Omega \subset \mathbb{R}^n$  is a bounded domain with piecewise  $C^1$  boundary condition that admits Green's function  $G_y$ . Then, for  $f \in C^0(\Omega)$  and  $g \in C^0(\partial\Omega)$ ,

$$\begin{aligned} -\Delta u &= f \\ u|_{\partial\Omega} &= g \end{aligned}$$

is solved by

$$u(y) = -\int_{\Omega} f G_y dx - \int_{\partial\Omega} g \partial_\eta G_y dS$$

*Proof:*

By Green's second identity,

$$\int_{\Omega} H_y \Delta u - u \Delta H_y dy = \int_{\partial\Omega} H_y \partial_\eta dS$$

Using the representation formula,

$$\begin{aligned} u(y) &= -\int_{\Omega} \Phi_y \Delta u dx + \int_{\partial\Omega} [\Phi_y \partial_\eta u - u \partial_\eta \Phi_y] dS - 0 \\ &= -\int_{\Omega} \Phi_y \Delta u dx + \int_{\partial\Omega} [\Phi_y \partial_\eta u - u \partial_\eta \Phi_y] dS - \int_{\Omega} H_y \Delta u dy + \int_{\partial\Omega} H_y \partial_\eta dS \end{aligned}$$

which, combining terms, is the formula above.

*Q.E.D.*

The method of images

**Example:** The Half-Plane

**Example:** The Unit Ball

## 4 Time-Dependent Fundamental Solutions

[Wave, Heat, Schrodinger]